

Tomáš Šoltinský

Curriculum Vitae

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Employment

2024
2026

Postdoctoral Research Grant Holder, *National Institute of Astrophysics - Astronomical Observatory of Trieste*, Trieste, Italy

2023-24

Postdoctoral Research Fellow, *Tata Institute of Fundamental Research*, Mumbai, India
Department of Theoretical Physics

Education

2019-23

PhD in Physics, *The University of Nottingham*, Nottingham, UK

Supervisors: Dr James S Bolton, Dr Nina Hatch

Thesis: Simulating neutral hydrogen across cosmic time: the 21-cm forest in late reionization models and the global 21-cm signal at cosmic dawn

○ Vice-Chancellor's Scholarship for Research Excellence

○ Graduated July 2023

2014-19

MPhys in Physics, Astrophysics, *The University of Edinburgh*, Edinburgh, UK

○ Graduated July 2019 with the First Class

2016-17

Exchange program, *California Institute of Technology*, Pasadena, USA

○ GPA: 3.3

Observing Proposals

2026

PI of uGMRT Cycle 50 observations (ID: 50.010), *A uGMRT Band-2 Search for the 21-cm Forest 1D Power Spectrum at $z \simeq 5.8$* , 60 hr awarded

Software

2024

21cm-forest_1DPS repository, https://github.com/tomassoltinsky/21cm-forest_1DPS

Forward-modelling the 21-cm forest with instrumental features and compute its 1D power spectrum

Used in 2 publications

Technical and Academic Experience

2025

Refereeing scientific papers, for *A&A*, *PRL*, *PRD*

2023-24

In-Kind Directable Software Development for the LSST-DESC, *Tata Institute of Fundamental Research*, Mumbai, India

○ Development of a code for calculating 3D trispectrum from N-body simulations

2018-19

MPhys Project, *The University of Edinburgh*, Edinburgh, UK

Extreme Variability of Active Galactic Nuclei ([report](#))

Supervisor: Prof Andy Lawrence

○ Tested black hole accretion disc models in the context of extreme variability of AGN luminosity

○ Included a public summary (created a [card game](#))

2018

Cormack Vacation Research Scholarship and Deans' Vacation Scholarship, *The Institute for Astronomy*, Edinburgh, UK

21cm Signal Marking the End of Dark Ages ([report](#), [poster](#))

Supervisor: Dr José Oñorbe

○ Mock observations of global 21cm signal with focus on Dark Ages based on Nyx simulations

2018

Senior Honours Project, *The University of Edinburgh*, Edinburgh, UK

Scalar Field Dark Energy

Supervisor: Prof Andy Taylor

○ Simulated the expansion of the universe driven by dark energy modeled as a dynamical scalar field

2017-18

Telescope Group Project, *The University of Edinburgh*, Edinburgh, UK

Measurement of the Characterising Properties of Exoplanet TrES-2b and its Host Star

Supervisor: Prof Philip Best

○ Transit Photometry and Doppler Spectroscopy methods

○ Observations performed with Edinburgh University's Meade MAX 20in ACF telescope

2017

Summer Undergraduate Research Fellowship, Caltech, Pasadena, USA

Redefining the Habitable Zone

Supervisor: Prof David J. Stevenson

- Determining conditions under which water in liquid phase can be present on the surface of a planet

2017

Research in Experimental Physics, Caltech, LIGO, Pasadena, USA

Signal Processing

Supervisor: Dr Thomas J. Massinger

- Data analysis and investigation of glitches in data

Publications

T Šoltinský, G Autieri, V Iršič, et al., "The impact of the IGM thermal state on the Ly α flux 3D power spectrum from linear to highly non-linear scales", submitted to *JCAP*, 2026. ([link](#))

G Autieri, V Iršič, **T Šoltinský**, et al., "Physical Calibration of a Minimal Effective Field Theory of the Three-Dimensional Lyman- α Forest", submitted to *JCAP*, 2026. ([link](#))

T Šoltinský, A Chakraborty, G Kulkarni, et al., "Forest without Trees is still Fruitful: Constraints on the thermal state of the neutral IGM at $z \approx 5.6$ with the 21-cm forest power spectrum", submitted to *Nature Communications*, 2026. ([link](#))

CD Science Working Group, "Overview of 21cm Experiments at high redshift with SKAO", published in *Advancing Astrophysics with the SKAII (AASKAII)*, 2026 (arXiv:2606.20366). ([link](#))

J Cang, B Ciardi, B Maity, et al., "Exploring the Cosmic Dawn through the 21 cm Forest and High-redshift Radio Sources with the SKA", published in *Advancing Astrophysics with the SKAII (AASKAII)*, 2026 (arXiv:2606.20366). ([link](#))

S K Patil, **T Šoltinský**, S Maitra, et al., "Efficient neutral-IGM inference from noisy 21-cm forest spectra with latent-space U-Net encoding and XGBoost", published in *MNRAS*, Volume 546, Issue 4, id.stag236, 18 pp., 2026. ([link](#))

T Šoltinský, G Kulkarni, S P Tendulkar, et al., "Prospects of a statistical detection of the 21-cm forest and its potential to constrain the thermal state of the neutral IGM during reionization", published in *MNRAS*, Volume 537, Issue 1, pp.364-378, 2025. ([link](#))

E Puchwein, J S Bolton, L C Keating, et al. "The Sherwood-Relics simulations: overview and impact of patchy reionization and pressure smoothing on the intergalactic medium", published in *MNRAS*, Volume 519, Issue 4, pp.6162-6183, 2023. ([link](#))

T Šoltinský, J S Bolton, M Molaro, et al. "Probing quasar lifetimes with proximate 21-centimetre absorption in the diffuse intergalactic medium at redshifts $z \geq 6$ ", published in *MNRAS*, Volume 519, Issue 2, pp.3027-3045, 2023. ([link](#))

T Šoltinský, J S Bolton, N Hatch, et al. "The detectability of strong 21-cm forest absorbers from the diffuse intergalactic medium in late reionization models", published in *MNRAS*, Volume 506, pp. 5818-5835, 2021. ([link](#))

C S Cockell, "Astrobiology: Understanding Life in the Universe", *Wiley-Blackwell*, 2015

- Contributed to and was cited in the Preface to the book

Awards, Scholarship, Grants

2019-22

Vice Chancellor's Scholarship for Research Excellence (EU), *The University of Nottingham*, Nottingham, UK

To support my PhD studies

2019

Students into the World grant, *The Tatra banka Foundation*, Slovakia

2018

RSE Cormack Undergraduate Astronomy Research Prize, *Royal Society of Edinburgh*, Edinburgh, UK

For the "21cm Marking the End of Dark Ages" project

2016

Year 2 Physics Class medal, *The University of Edinburgh*, Edinburgh, UK

For excellent performance during the 2015/16 session

- 2014 ● **The memorial letter of St. Gorazd**, *The Ministry of Education, Science, Research and Sport of the Slovak Republic*, Bratislava, Slovakia
For successful representation of the Slovak Republic at Austrian Young Physicists' Tournament and achievement of bronze medal
- 2014 ● **Best in Natural Sciences school award**, *Pavol Horov Gymnasium*, Michalovce, Slovakia
For representation of the school at various Physics and Maths competitions

Mentoring and Teaching

- 2021 ● **Mentoring at GROWNi.sk**, Slovakia
○ Advising students in their career and study decisions
- 2016 ● **Outreach talks**, *Pavol Horov Gymnasium*, Michalovce, Slovakia
○ "From LA through half the globe to simulated universe", Topic: career path and simulations in astrophysics
○ "From LA to LA", Topic: gained experience and how to study abroad
○ "Science has no borders", Topic: my research experience and ongoing research in the world
- 2023 ● **Advanced 21-cm Cosmology School and Workshop**, *National Institute of Science Education and Research*, Bhubaneswar, India, Lectures ([link](#))

Service to Academic Community

- 2023-24 ● **State of the Universe Seminar Series Organizer**, *TIFR*, Mumbai, India
- 2022 ● **Astronomy Postgraduate Student Representative**, *The University of Nottingham*, Nottingham, UK
○ Main point of communication between PG students and staff
- 2021-22 ● **Lunch Talks Organizer**, *The University of Nottingham*, Nottingham, UK
- 2019-22 ● **Physics Coursework Marker**, *The University of Nottingham*, Nottingham, UK
- 2017-21 ● **Student Ambassador**, *The University of Edinburgh and the University of Nottingham*, Edinburgh and Nottingham, UK
○ Resided over the information desk and gave tours of the campus and the School of Physics and Astronomy at the Open Days, Post-offer Visit Days, Visiting Students Welcome Event
- 2018 ● **UUKi Parliamentary Reception for the Go International: Stand Out Campaign**, *Westminster*, London, UK
○ Ambassador for The University of Edinburgh

Presentations at International Conferences and Meetings

- 2026 ● **EAS 2026**, *Swisstech convention centre*, Lausanne, Switzerland, Talks and posters at 3 sessions ([link 1](#), [link 2](#))
- 2026 ● **Cosmology Seminar**, *The Institut d'Astrophysique Spatiale*, Orsay-Paris, France, Seminar
- 2026 ● **Cosmology Journal Club**, *University of Padova*, Padova, Italy, Seminar
- 2025 ● **The Fifth National Workshop on the SKA Project - From precursors to SKAO: shaping the future of Italian radio astronomy**, *CNR*, Bologna, Italy, Talk
- 2025 ● **Kaba Kada: Exploring the first billion years of the Universe**, *Oaks Port Douglas Resort*, Port Douglas, Queensland, Invited talk and part of panel
- 2025 ● **ICRAR-Curtin Colloquium**, *Curtin University*, Perth, Western Australia, Colloquium
- 2025 ● **NAM 2025**, *Durham University*, Durham, UK, Talk and a poster at 2 sessions
- 2025 ● **The galaxy-IGM connection in the first billion years**, *IFPU*, Trieste, Italy, Talk
- 2025 ● **Radio Cosmology and Continuum Observations in the SKA Era: A Synergic View**, *International Centre for Theoretical Sciences*, Bengaluru, India, Talk ([link](#))
- 2024 ● **SKA Cosmology SWG Meeting**, *Observatoire de la Cote d'Azur*, Nice, France, Talk
- 2024 ● **IFPU colloquium**, *IFPU*, Trieste, Italy, Colloquium
- 2024 ● **The First Gigayear(s)**, *Hilo Hawaiian Hotel*, Hilo, Hawai'i, Talk
- 2024 ● **The Origin and Evolution of Supermassive Black Holes**, *Sexten Center for Astrophysics*, Sexten, Italy, Talk ([link 1](#), [link 2](#))

2024	Cosmic Dawn at High Latitudes Conference , <i>Royal Swedish Academy of Sciences</i> , Stockholm, Sweden, Talk (link)
2024	State of the Universe , <i>TIFR</i> , Mumbai, India, Seminar (link)
2024	Cosmology in the Alps Conference , <i>Les Diablerets</i> , Switzerland, Talk (link)
2024	Monday Afternoon Talks , <i>MIT Kavli Institute</i> , Cambridge, US, Talk
2024	Pune-Mumbai Cosmology and Astro-Particle meeting , <i>The Inter-University Centre for Astronomy and Astrophysics</i> , Pune, India, Talk and Chaired a session (link)
2024	Astronomical Society of India meeting , <i>IISc</i> , Bengaluru, India, Talk
2024	HEA group meeting , <i>Masaryk University</i> , Brno, Czechia, Seminar
2023	Astrophysics Seminar , <i>University of Helsinki</i> , Helsinki, Finland, Seminar
2023	Galaxy coffee , <i>Max Planck Institute for Astronomy</i> , Heidelberg, Germany, Talk
2023	21-cm Series Seminar , <i>University of Cambridge</i> , Cambridge, UK, Seminar
2023	Colloquium , <i>National Centre for Radio Astrophysics</i> , Pune, India, Colloquium
2022	Lunch talk , <i>The University of Nottingham</i> , Nottingham, UK, Talk
2022	21-cm Series Seminar , <i>University of Cambridge</i> , Cambridge, UK, Seminar
2021	State of the Universe , <i>TIFR</i> , Mumbai, India, Seminar (link)
2021	NAM 2021 , <i>The University of Bath</i> , Bath, UK, Talk
2021	SAZERAC 2.0 , Recorded talk (link)
2021	Lunch talk , <i>The University of Nottingham</i> , Nottingham, UK, Talk
2020	Lunch talk , <i>The University of Nottingham</i> , Nottingham, UK, Talk

Membership in collaborations

2024	SKA Cosmology and Epoch of Reionization SWGs , link Cosmology , link EoR
2024	21-cm Cosmology Group (India) , link
2023	LSST-DESC , Large scale structure WG , provisional membership, link
2019	Sherwood-Relics simulation suite , link

Training/Skills

2023	Foundation HPC-Skills Course , <i>DiRAC</i>
2023	DP0 Virtual Summer School 2023 , <i>Vera C. Rubin Observatory/LSST</i>
Programming	Python, Jupyter notebook, C, $\text{\LaTeX}$, Mathematica, IDL, Radiative Transfer, 21cmFAST, IRAF, SExtractor, Large data analysis, MCMC, OpenMP parallelisation, Slurm